Fiscal Year:	FY 2019	**Task Last Updated:**	FY 03/18/2019
PI Name:	Snell, Edward H. Ph.D.		
Project Title:	Growth Rate Dispersion as a Predictive Indicator for Biological Crystal Samples Where Quality Can be Improved with Microgravity Growth		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	BIOPHYSICS--Biophysics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	esnell@hwi.buffalo.edu	**Fax:**	FY
PI Organization Type:	NON-PROFIT	**Phone:**	716-898-8631
Organization Name:	Hauptman-Woodward Medical Research Institute		
PI Address 1:	Dept. of Structural Biology		
PI Address 2:	700 Ellicott Street		
PI Web Page:			
City:	Buffalo	**State:**	NY
Zip Code:	14203-1102	**Congressional District:**	26
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2013 Complex Fluids & Macromolecular Biophysics NNH13ZTT001N
Start Date:	01/24/2014	**End Date:**	01/23/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Penn, Benjamin	**Contact Phone:**	
Contact Email:	benjamin.g.penndr@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Grant extends to 1/23/2021 with new grant number 80NSSC19K1665; information per B. Penn/MSFC (Ed., 11/12/19) NOTE: End date changed to 1/23/2020 per NSSC information (Ed., 3/12/19)		
Key Personnel Changes/Previous PI:	November 2018 report: Tim Stachowski is a student at Roswell Park Comprehensive Cancer Center. Now that the initial flight samples have returned and X-ray analysis has begun Tim has been involved with the X-ray analysis and developing samples for the next flight. His samples are now on orbit as part of LMMBIO-6.		
COI Name (Institution):			
Grant/Contract No.:	80NSSC19K1665 ; NNX14AD80G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	EDITOR's NOTE (11/12/2019): Grant extends to 1/23/2021 with new grant number 80NSSC19K1665; information per B. Penn/MSFC. Biological macromolecules make up the machinery, the instruction set, and the scaffold of life. They are smaller than the wavelength of visible light; thus, sophisticated techniques are needed to visualize them, and through this process, understand how life works. Structural understanding helps to discern mechanisms; once we understand how the machinery works, we aid, or more commonly impede that machinery through appropriate pharmaceutical design. The principal means to visualize the structure of these macromolecules is X-ray crystallography, a process that measures the diffraction of X-rays from ordered crystals to calculate the atomic structure. The importance of this process is exemplified in over 12 Nobel Prizes awarded to discoveries made from biological structures derived by X-ray crystallography. A limiting element to the process is the availability of high-quality, well-diffracting crystals. Quality is defined as a crystal that provides X-ray diffraction data of sufficient completeness and detail (resolution) to see the structure and hence understand the biology of the system. Biological macromolecules often take many days to grow crystals. Growth in reduced acceleration (commonly termed microgravity) on an orbiting spacecraft extends the physical quality of macromolecular crystals through a reduction in the mosaic spread (caused by slight misalignments of the molecules in the crystal lattice and used as a measure of long-range order) and an increase in crystal volume. Resolution (a measure of short-range order) is not directly affected by microgravity, but can still benefit with the correct experimental design to exploit the demonstrated improvements in long-range order. Microgravity growth can yield an improved quality crystal; however, not all samples improve from microgravity growth. If we could predict which samples could be improved by crystal growth in microgravity, then the true potential of this medium could be exploited in an efficient manner. It is our hypothesis that growth rate dispersion could be an indicator of crystals whose quality could be improved in microgravity. Growth rate dispersion is a phenomenon encountered in crystallization where seemingly identical crystals, produced from the same conditions, grow at different rates. We contend that large growth rate dispersion on the ground is indicative of a sample that should be improved by microgravity growth. In the case of inorganic crystals, a decrease in growth rate dispersion is correlated with an increase in physical crystal quality. The dispersion itself is thought to be caused by surface artifacts on the growing crystals that induce lattice imperfections. Gravity enables convective flows to develop in a crystallization solution. These flows are driven by density differences as molecules are depleted from the surrounding solution during nucleation and growth of the crystal. We plan to monitor growth rate dispersion of crystals grown on the ground and in microgravity, to determine if there is a correlation between the physical qualities of the resulting crystals with those measurements. We propose to use molecular biology techniques to alter the crystallization contacts and shift the growth rate dispersion properties of a single protein from low to high to test our predictive hypothesis. Finally we propose to extend the study to a selection of good and poorly diffracting crystals on the ground and confirm that those displaying high-growth rate dispersion on the ground are those that are improved on orbit and generate improved structural data when their quality is exploited.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	If our hypothesis proves to be correct, we will develop a generally applicable means to identify macromolecular crystals where quality will improve with microgravity growth. Biological macromolecules make up the machinery, the instruction set, and the scaffold of life. They are smaller than the wavelength of visible light; thus, sophisticated techniques are needed to visualize them, and through this process, understand how life works. Structural understanding helps to discern mechanisms; once we understand how the machinery works, we aid, or more commonly impede that machinery through appropriate pharmaceutical design. The principal means to visualize the structure of these macromolecules is X-ray crystallography, which accounts for 90% of the known structural information. It is a process that measures the diffraction of X-rays from ordered crystals to calculate the atomic structure. The importance of this process is exemplified by 12 Nobel Prizes awarded to discoveries made from X-ray crystallographic structures of biological macromolecules. Crystallography requires high-quality, well-diffracting crystals; crystallization is the most significant impediment to structure determination. Quality is defined as a crystal that provides X-ray diffraction data of sufficient completeness and detail (resolution) to see the structure and hence understand the biology of the system. In our high-throughput crystallization screening laboratory at the Hauptman-Woodward Medical Research Institute approximately 50% of proteins samples crystallize in at least one chemical condition tested and of those, approximately 50% go on to be deposited within the protein data bank – a depository for atomic resolution structural data. To restate this, some 25% of the macromolecules that enter the laboratory provide crystals that are of insufficient quality for structural determination, and for the 25% of macromolecules that do provide a structure, higher-quality crystals would enhance the structural detail to enable a better understanding of the molecular function. Microgravity crystallization has the potential to improve crystal quality and thereby provide novel structures for the 25% of macromolecules that produced crystals of insufficient quality for structure determination. In addition, it has the potential to improve the level of detail that can be observed in structures that have previously been resolved. It has become increasingly apparent that many of the failures to get structural data from crystals are not that they don't diffract, but that the diffraction is to very low resolution. A more ordered, physically higher quality crystal would increase the signal-to-noise of recorded data and extend this resolution to a point where structural information may be available, or potential pathways for improvement could be quantitatively evaluated.
Task Progress:	Crystals of all T4 lysozyme samples have been produced on orbit. Biologically important samples are now on orbit with crystallization be activated in 2019. A book has been published by the Principal Investigator (PI), Biological Small Angle X-ray Scattering. by Lattman, Grant and Snell, Oxford University Press. A paper has been published by the PI in collaboration with partners including large pharma companies and GoogleBrain - Classification of crystallization outcomes using deep convolutional neural networks. Bruno AE, Charbonneau P, Newman J, Snell EH, So DR, Vanhoucke V, Watkins CJ, Williams S, Wilson J. PLoS One. 2018 Jun 20;13(6):e0198883. doi: 10.1371/journal.pone.019888 Another paper is in press involving ground experiments on one of the proteins currently on orbit. Details of this will be reported on publication.

	[Ed. note: see also Bibliography section below]
Bibliography Type:	Description: (Last Updated: 12/14/2021)
Articles in Peer-reviewed Journals	Bruno AE, Charbonneau P, Newman J, Snell EH, So DR, Vanhoucke V, Watkins CJ, Williams S, Wilson J. "Classification of crystallization outcomes using deep convolutional neural networks." PLoS One. 2018 Jun 20;13(6):e0198883. eCollection 2018. https://doi.org/10.1371/journal.pone.0198883 ; PubMed PMID: 29924841; PubMed Central PMCID: PMC6010233 , Jun-2018
Articles in Peer-reviewed Journals	Niessen KA, Xu M, George DK, Chen MC, Ferré-D'Amaré AR, Snell EH, Cody V, Pace J, Schmidt M, Markelz AG. "Protein and RNA dynamical fingerprinting." Nat Commun. 2019 Mar 4;10(1):1026. https://doi.org/10.1038/s41467-019-08926-3 ; PubMed PMID: 30833555; PubMed Central PMCID: PMC6399446 , Mar-2019
Books/Book Chapters	Lattman EE, Grant TD, Snell EH. "Biological Small Angle Scattering: Theory and Practice." Oxford: Oxford University Press, 2018. IUCr monographs on crystallography, 29. https://doi.org/10.1093/oso/9780199670871.001.0001 , Sep-2018